

BITAN PAUL

AI / ML & DATA ENGINEER • GenAI • RAG • LLMs • AGENTIC SYSTEMS

Bengaluru, India | +91 80179 63283 | thebitanpaul@gmail.com | [linkedin.com/in/thebitanpaul](https://www.linkedin.com/in/thebitanpaul) | github.com/thebitanpaul | phiuture.com

PROFESSIONAL SUMMARY

AI/ML & Data Engineer with 2+ years building production-grade Generative AI, RAG, and agentic systems. Hands-on across RAG and multi-agent LLM orchestration, computer vision, and lakehouse data engineering with validation/data-quality automation on AWS, Azure, Snowflake, Databricks, Airflow, and Kafka. Ships scalable, well-tested Python systems with CI/CD.

TECHNICAL SKILLS

Languages: Python, Java, SQL

GenAI & ML: LLMs, RAG (Retrieval-Augmented Generation), Hybrid Retrieval (BM25 + Embeddings), Vector Search (FAISS), Agentic AI, Multi-Agent Systems, LangChain, Prompt Engineering, LLM APIs, Deep Learning, Computer Vision (CNN), NLP, TensorFlow, Keras

Data Engineering: Snowflake (Cortex Analyst, Semantic Models, governed Text-to-SQL), Databricks, Delta Lake, Unity Catalog, PySpark, Apache Airflow, Apache Kafka, Medallion (Lakehouse) Architecture, ETL & Streaming, Streamlit, Plotly, Time-Series Forecasting (SARIMA, MSTL, Prophet, LogOP)

Cloud & Backend: AWS (Bedrock, Lambda, S3, CloudFront, ECR, CloudWatch), Azure (AI Foundry, DevOps, ADLS, Key Vault), FastAPI, REST APIs, WebSockets, JWT Auth, MySQL, Firebase

AI-Assisted Dev: Claude Code CLI, AI-assisted dev with rigorous pre-ship code validation; React, TypeScript, Vite, Tailwind CSS, shadcn/ui

Testing & Validation: pytest, Test Automation, Great Expectations (Data Quality), DQ Gating & Quarantine, DAG Integrity Testing, Deterministic & Claim Verification, Pre-Deployment Checks

Tools & DevOps: Git/GitHub, CI/CD, Docker, Azure DevOps, Vercel, Render, Android Studio, Google Play Console

PROFESSIONAL EXPERIENCE

Navikenz — Data Engineering & AI Specialist

Jan 2024 – Present

Bengaluru, India • Hybrid

Global Multinational Law Firm — AI Contract Intelligence Platform

Client

- Architected a **FastAPI + React/TypeScript** platform automating contract gap analysis via chained **LLM agents (Amazon Bedrock)** over a **FAISS** vector store; added real-time **WebSocket** collaboration (multi-user editing, live cursors) on a **MySQL** store with clause versioning, **JWT** + **bcrypt** auth and an **AWS S3** ingestion pipeline with client-side **DOCX/PDF** export.

Global Investment Management Firm — Conversational BI Platform (Snowflake Cortex)

Client

- Built an end-to-end conversational BI platform on **Snowflake Cortex Analyst** that turns natural-language questions into governed SQL over investment data — authored the **semantic model** (fact/dimension tables, synonyms, verified queries) and shipped a **Streamlit-in-Snowflake** app with auto-selected **Plotly** charts, business narratives, SQL transparency, and confidence validation, plus investor-grade analytics (P&L, risk, drawdown, CAGR).

Global Multinational Pharmaceutical Enterprise — Azure Data Engineering & Databricks

Client

- Built and maintained **Databricks pipelines with Unity Catalog** governance, automated scheduling, and **LangChain**-based UDFs; integrated **Azure Storage, Key Vaults, ADLS, and Graph/Management APIs** for secure, scalable data handling.

Multinational Healthcare Provider — Time-Series Forecasting & EDA

Client

- Benchmarked **SARIMA, MSTL, and Prophet** on large-scale time-series data, combining outputs via weighted averaging and the **logarithmic opinion pool (LogOP)** ensemble to significantly improve forecast accuracy.

Internal AI Tooling — NDLT & GenBI

Internal

- Built **NDLT** (format-preserving document translator) and **GenBI** — GenAI BI APIs on **Azure AI Foundry** with Azure ETL pipelines and CI/CD via **Azure DevOps**.

Outlier AI — AI Trainer (Freelance, Remote)

Sep 2024 – Oct 2025

- Refined LLMs on reasoning, coding, and NLP tasks; specialized in **prompt engineering, error analysis, and model-behavior optimization**.

KEY PROJECTS

VERA — Version-Explicit Retrieval & Answering (RAG for Regulated Clinical Workflows)

[Live Demo](#) · [GitHub](#)

- Built a **RAG system** for clinical-trial protocols using deterministic version resolution and **hybrid retrieval (BM25 + embeddings)**, with an **agentic verifier** that cites sources and **safe-abstains/escalates** when evidence is insufficient; **4-provider bring-your-own-key backend**; **135 automated pytest tests**.

ArchAltect — AI Microservice Architecture Assistant

[Live Demo](#) · [GitHub](#)

- Built a **multi-agent app (FastAPI + React)** that turns a requirements document into a full microservice architecture — service map, blast-radius resilience simulation, metrics, risk audit, and migration roadmap. Engineered a token-efficient **“logic-over-LLM”** pipeline (pure-Python analyses at 0 tokens) across **10 LLM providers** with encrypted bring-your-own-key.

DataNexus — E-Commerce Data Lakehouse & Orchestration Pipeline

[GitHub](#)

- Architected a production-style **medallion lakehouse** (Bronze→Silver→Gold) on **Databricks/PySpark**, orchestrated by **Apache Airflow** behind a **Great Expectations DQ gate** with quarantine, plus an **Apache Kafka** streaming path and automated DAG-integrity tests.

phiUture — Portfolio & Product Showcase

phiuture.com

- Shipped a **production-grade portfolio site (React, TypeScript, Tailwind, shadcn/ui)** in **5 days with Claude Code CLI** — **validating all AI-generated code before shipping**; deployed on **Vercel** with full technical SEO (JSON-LD, OpenGraph, IIm.txt).

Campus Buddy — Android App (Published)

[Play Store](#)

- Shipped a published Android app (**Firebase, Power Automate**) solving real campus problems — **1,000+ downloads, 600+ active users, 4.8★ STARS**; the project secured my role at Navikenz.

Face Says BMI — Deep Learning / Computer Vision

[GitHub](#)

- Built a **CNN** (MT-CNN, OpenCV, VGGFace, Keras/TensorFlow) predicting BMI, age, and gender from a single facial image.

EDUCATION & CERTIFICATIONS

B.E. in Computer Science — AI & Machine Learning Specialization — Chandigarh University, CGPA 8.04 2020 – 2024

Certifications: ML to Deep Learning — ISRO/IIRS (2022) • SQL — HackerRank (2023) • Apache Kafka — Udemey (2026)